

Question	Marking Guidance	Mark	Comments
06.1	(Genome) 1. Complete set of genes in a cell OR (All) the DNA in a cell OR (All) the genes/alleles/genetic material in a cell OR The total number of DNA bases in a cell; (Proteome) 2. (Full) range of proteins that a cell can produce OR (Full) range of proteins coded for by the cell's DNA/genome;	2 (2 x AO1)	1. Reject 'all the DNA/genes within a species/population' 1 and 2. Accept organism for cell 2. For 'full range' accept 'complete set' OR all 2. Do not accept 'number of proteins' unqualified 2. Ignore 'range of proteins that a species/population can produce'

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06.2	1. Can not identify/distinguish species; 2. (Optical) microscope resolution is low; 3. Flagella (are fragile so) broken/damaged/missing OR Artefacts misinterpreted/mistaken for flagella; 4. (Flagella) difficult to stain so not visible; 5. (Flagella) at an angle so not visible; 6. Not all bacteria have flagella;	3 max (3 x AO3)	2. Ignore magnification 4. Ignore difficult to stain unqualified 5. Accept 'out of plane' for 'at an angle'

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06.3	DNA/mRNA/RNA <u>base</u> sequencing OR Amino acid sequencing OR Use of electron microscopes with greater resolution OR Use of electron microscopes and improved staining/preparation;	1 (AO1)	Accept genome sequencing Ignore detail OR magnification for resolution Accept abbreviations TEM OR SEM in this instance

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06.4	Correct answer of 19 565 (from measurement of 45mm) OR 20 000 (from measurement of 46mm) OR 20 435 (from measurement of 47mm) = 2 marks ;; Accept for 1 mark , evidence of 45 000 OR 46 000 OR 47 000 (correct image size in μm) OR $\div 2.3$ (correct use of equation);	2 (2 x AO2)	Accept answers that round correctly to those shown Accept division by 2.3×10^{-3} OR 0.0023
